



EEG characteristics predict subsequent epilepsy in children with their first unprovoked seizure

Hideaki Kanemura^{a,*}, Fumikazu Sano^a, Tetsuo Ohyama^a, Sonoko Mizorogi^a, Kanji Sugita^a, Masao Aihara^b

^a Department of Pediatrics, Faculty of Medicine, University of Yamanashi, 1110 Shimokato, Chuo, Yamanashi 409-3898, Japan

^b Interdisciplinary Graduate School of Medicine and Engineering, University of Yamanashi, 1110 Shimokato, Chuo, Yamanashi 409-3898, Japan

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ABSTRACT

One of the most important risk factors for seizure recurrence is the electroencephalogram (EEG) characteristics of children with a first unprovoked seizure. However, the nature of the relationship between the risk factors for subsequent epilepsy and EEG characteristics, especially the localization of EEG paroxysmal abnormalities (PAs), remains unclear. The importance of EEG characteristics, especially the localizations of PAs, as predictors for subsequent epilepsy in children with a first unprovoked seizure was investigated. The participants were recruited from University of Yamanashi Hospital between July 1, 1997 and June 30, 2010 and followed until December 31, 2014. Eligible candidates were children between 1 month and 15 years old who presented with their first unprovoked non-febrile seizure. Awake and sleep EEGs were performed on a 12- or 16-channel machine 7–20 days after a first unprovoked seizure. Agreement regarding the presence of a PA was required for inclusion of the patient in the study. EEG PAs were classified based on the presence of spikes, sharp waves, or spike-wave complexes, whether focal or generalized, that were considered abnormal for age and state. All subjects were followed for more than 4 years. Of 87 subjects, 48 (55.2%) experienced recurrence. On the other hand, of 87 subjects, 52 (59.8%) showed an EEG PA. Of 18 patients with paroxysms in the frontal region, 17 (94.4%) developed epilepsy. Patients with frontal EEG paroxysms had a significantly higher risk of developing epilepsy than those with focal paroxysms in other regions of EEG foci ($p < 0.05$). Moreover, compared with generalized EEG foci, the odds ratios for patients with frontal, RD, occipital, and mid-temporal EEG foci were 85.0 (95% CI: 4.5–1617.1), 9.3 (0.9–96.0), 2.5 (0.1–62.6), and 7.5 (0.5–122.7), respectively. Children with a first unprovoked seizure presenting with frontal paroxysmal EEG abnormalities may be at risk for the development of epilepsy later in life.

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1. Introduction

A first unprovoked seizure almost automatically generates the need for a number of diagnostic investigations, raises numerous questions about evolution and prognosis, and necessitates a treatment or no treatment decision. The therapeutic decisions to follow have important implications for health and quality of life (QOL). The physicians have to address several clinical questions, including the impact of treatment on the risk of seizure recurrence and on the

long-term prognosis of epilepsy. In addition, the risk-benefit ratio of pharmacological treatment should also be considered. Treatment for most patients is with antiepileptic drugs (AEDs), which carry risks of acute idiosyncratic reactions, dose-related and chronic toxic effects, and teratogenicity. Thus, any exploration of the approach to a first unprovoked seizure requires an understanding of the natural history and prognosis of the seizure disorder, including, to a large degree, risk factors for recurrence and the development of epilepsy (Haut et al., 2007).

Among the potential risk factors for seizure recurrence, the electroencephalogram (EEG) has been considered to be important. In particular, the EEG is an extremely important predictor of recurrence in patients with a non-remote symptomatic seizure and in children (Berg and Shinnar, 1991; Hirtz et al., 2000; Kim et al., 2006). Various studies of recurrence risk following a first unprovoked seizure in childhood have reported that the presence of an abnormal EEG confers a higher recurrence risk than a

Abbreviations: PA, paroxysmal abnormality; RD, rolandic discharge.

* Corresponding author. Tel.: +81 55 273 9606; fax: +81 55 273 6745.

E-mail addresses: ykimu@yamanashi.ac.jp (H. Kanemura),

fsano@yamanashi.ac.jp (F. Sano), tohyama@yamanashi.ac.jp (T. Ohyama),

sonokoh@yamanashi.ac.jp (S. Mizorogi), ksugita@yamanashi.ac.jp (K. Sugita),

maihara@yamanashi.ac.jp (M. Aihara).

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normal EEG (Berg and Shinnar, 1991; Hirtz et al., 2000; Shinnar et al., 2000). Thus, an EEG is considered a standard part of the diagnostic evaluation of a first unprovoked seizure in children (Kim et al., 2006). Although EEG abnormalities may be present in these children, their clinical significance is unclear. Moreover, the nature of the relationships between the risk factors for subsequent epilepsy and EEG characteristics, especially the localization of EEG paroxysmal abnormalities (PAs), also remains unclear.

The literature on seizure recurrence in pediatric patients is widely inconsistent, depending on the follow-up period and the population studied (Berg and Shinnar, 1991; Hirtz et al., 2000; Shinnar et al., 2000; Winckler and Rotta, 2004; Mizorogi et al., 2015). We have followed 87 children from the time of their first unprovoked seizure. The present analysis focuses on the risk of subsequent recurrences after an initial seizure recurrence and risk factors, especially EEG PA localization, for further recurrences.

2. Materials and methods

2.1. Participants

Participants were recruited from the University of Yamanashi Hospital between July 1, 1997 and June 30, 2010 and followed until December 31, 2014. The records of each participant were reviewed to document the occurrence of a seizure, classify the cause, and verify residency. Eligible candidates were children between 1 month and 15 years old who presented with their first unprovoked non-febrile seizure. All participants were required to agree to take no medication until the appearance of a second seizure. Participants with multiple seizures occurring within 24 h were included if they were not taking AEDs. Participants with previous neonatal seizures, febrile convulsions, immediate posttraumatic seizures, or other provoked seizures who now presented with a first unprovoked seizure were also included. Exclusion criteria were ambiguous episodes, seizures occurring with acute encephalopathy (for example, meningitis, encephalitis, or electrolyte or glucose disturbances, within 1 week of significant head trauma, or in association with febrile illness), or the presence of organic brain lesions on magnetic resonance imaging (MRI), such as tumor, brain injuries, progressive neurological diseases, or metabolic diseases. Furthermore, children with status epilepticus (defined as a seizure lasting >30 min or a series of seizures lasting >30 min without the child regaining consciousness) were also excluded because of the need to give AEDs and the tendency to then continue AED treatment. EEGs were scheduled for all patients. Patients were referred from the outpatient department for an EEG within 7–20 days after the seizure. All patients were evaluated on referral with a standardized history and neurologic examination. Etiology and seizure type were classified in accordance with International League Against Epilepsy criteria (Commission on Classification and Terminology of the International League Against Epilepsy, 1981; Commission on Epidemiology and Prognosis, International League Against Epilepsy, 1993). Epilepsy was considered present when unprovoked seizure recurrence was documented. A seizure was considered unprovoked when there was no identifiable proximate insult (e.g., fever or head trauma) that could account for the seizure (Shinnar et al., 1996).

2.2. EEG procedures

Awake and sleep EEGs were performed on a 12- or 16-channel machine 7–20 days after a first unprovoked seizure. The duration of tracings was at least 20 min. Monopolar and bipolar electroencephalographic recordings were obtained with the 10–20 international system of electrode placement. For inclusion, it was

necessary that at least one EEG be obtained without drug induction, showing a clear sequence of awake-drowsy-sleep-arousal-awake states. For this reason, parents were instructed to keep their children awake the night before the visit. Intermittent photic activation was done routinely, and hyperventilation was used when age permitted.

EEG studies were coded by number and read independently by two pediatric epileptologists or neurologists blinded to the identity of the patients. Agreement regarding the presence of a PA was required for inclusion of the patient in the study. If the neurologists disagreed about the major classification, a third opinion was obtained from an experienced reader. Recordings included sleep in all of the patients. Tracings were classified as: normal; with focal or generalized PA; and with unspecific (nonepileptiform) baseline abnormalities (Winckler and Rotta, 2004). EEG PAs were classified based on the presence of spikes, sharp waves, or spike-wave complexes, whether focal or generalized, that were considered abnormal for age and state. If the localization of a paroxysmal discharge had migrated, the area affected most was considered the localization of the EEG PA. All patients were evaluated according to their EEG findings and whether they were diagnosed with epilepsy during follow-up. The data were collected prospectively until the appearance of a second seizure. All subjects were followed for more than 4 years.

The study was carried out in accordance with the Declaration of Helsinki. Informed consent was obtained from the parents of each patient following a full explanation of the procedures to be undertaken.

2.3. Statistical analysis

Statistical analysis was performed with SPSS II software for descriptive analyses. Differences were evaluated using the Mann–Whitney *U* test, Chi-square test, Fisher's exact test, and Bonferroni corrections for the relationship between later development of epilepsy and localization of EEG PA. The odds ratio as estimated from the logistic model was used as a measure of association and referred to as relative risk.

3. Results

A total of 96 children met all entry criteria and underwent EEGs. Nine were excluded because no follow-up was obtained after the first unprovoked seizure. Data from 87 children (age range, 3 months–13.4 years) were analyzed. The clinical profiles of all patients are shown in Table 1. Of 87 subjects, 48 (55.2%) experienced seizure recurrence. On the other hand, of 87 subjects, 52 (59.8%) showed EEG PAs. The clinical profiles of the patients with respect to the presence of EEG PAs are shown in Table 2. In this study, there were 13 children with no EEG PAs but with epilepsy and 17 children with EEG PAs but no subsequent seizure recurrence. The sensitivity and specificity in this study were 72.9% and 56.4%, respectively. The positive likelihood ratio (LR) and negative LR were 1.67 (95% confidence interval [CI], 1.13–2.49) and 0.48 (95% CI, 0.28–0.82), respectively. The patients with EEG PAs showed a significantly higher rate of epilepsy than patients without ($p=0.008$). In addition, the seizure characteristic of focal seizure was also significantly higher in patients with EEG PAs than in patients without EEG PAs ($p=0.042$). Other manifestations, such as age at seizure onset, family history of epilepsy, previous history of febrile seizure (FS), and evidence of prior neurological abnormalities such as cerebral palsy, did not differ between the patients with and without EEG PAs. By contrast, the duration of the first unprovoked seizure was significantly longer in patients without EEG PAs than in patients with EEG PAs ($p=0.0014$). The clinical profile of the patients with

Table 1
Clinical profiles of the patients in this study.

	Without epilepsy (n = 39)	With epilepsy (n = 48)	p-Value
Age at onset (mean, years)	3.2	3.6	0.567
Sex (male, n)	22	25	0.829
History of perinatal events (n)	4	4	0.999
FH of epilepsy (n)	7	9	0.999
Developmental delay prior to seizure (n)	4	3	0.695
Seizure characteristics (n)			
Generalized	28	17	
Focal	11	31	0.001
Neurological abnormalities (n)	9	5	0.145
EEG PA (n)	17	35	0.008
Generalized	5	1	
Focal	12	34	0.012

EEG, electroencephalogram; PA, paroxysmal abnormality; FS, febrile seizure; FH, family history.

Table 2
Clinical profiles of the patients by the presence of EEG PAs.

	Without EEG PA (n = 35)	With EEG PA (n = 52)	p-Value
Age at first unprovoked seizure (mean, years)	3.3	3.9	0.536
PH of FS (n)	6	5	0.338
FH of FS (n)	4	3	0.432
FH of epilepsy (n)	7	9	0.984
Neurological abnormalities (n)	9	5	0.338
Seizure characteristics			
Generalized (n)	25	20	
Focal (n)	10	32	0.042
Duration (mean, min)	3.3	2.6	0.0014
Later development of epilepsy (n)	13	35	0.002

EEG, electroencephalogram; PA, paroxysmal abnormality; PH, past history; FS, febrile seizure; FH, family history.

EEG PAs is shown in Table 3. Of 52 patients with EEG PAs, 35 (67.3%) developed epilepsy. The frequency of the seizure characteristic of focal seizure and age at seizure onset were significantly higher in patients with later development of epilepsy ($p = 0.0019$, 0.034 , respectively). Other manifestations, such as family history of epilepsy, previous history of FS, evidence of prior neurological abnormalities, and duration of first unprovoked seizure did not differ between the epilepsy and non-epilepsy groups.

Associations between the localizations of paroxysmal discharges and later development of epilepsy are shown in Table 4. Of 6 patients with generalized paroxysmal spike and wave activity, 1 developed epilepsy. Of 20 patients with rolandic discharge (RD), 13 (65.0%) developed epilepsy. Of 18 patients with paroxysms in the frontal region, 17 (94.4%) developed epilepsy. Of 3 patients with paroxysms in the occipital region, 1 developed epilepsy. Of 5 patients with mid-temporal region paroxysms, 3 developed epilepsy. Patients with frontal EEG PAs had a significantly

Table 3
Clinical profiles of the patients presenting with EEG PAs.

	Without epilepsy (n = 17)	With epilepsy (n = 35)	p-Value
Age at first unprovoked seizure (mean, years)	2.6	4.6	0.038
PH of FS (n)	3	2	0.315
FH of FS (n)	2	1	0.246
FH of epilepsy (n)	4	5	0.451
Neurological abnormalities (n)	3	2	0.315
Seizure characteristics			
Generalized (n)	12	8	
Focal (n)	5	27	0.0019
Duration (mean, min)	2.7	2.5	0.545

PH, past history; FS, febrile seizure; FH, family history.

higher risk of developing epilepsy than those with focal paroxysms in other regions of EEG foci ($p < 0.05$). Moreover, compared with generalized EEG foci, the odds ratios for patients with frontal, RD, occipital, and mid-temporal EEG foci were 85.0 (95% CI, 4.5–1617.1), 9.3 (0.9–96.0), 2.5 (0.1–62.6), and 7.5 (0.5–122.7), respectively. Patients with frontal EEG PAs had a significantly higher risk of developing epilepsy than those with EEG PAs in other regions ($p = 0.0022$). In contrast, the mean intervals between the first and second seizures were not significantly different by the regions of the EEG PAs.

Associations between the EEG findings and the classification of subsequent epilepsy into epileptic syndromes were as follows. One patient with generalized PA was diagnosed as having generalized epilepsy with tonic-clonic seizures (GETCS). Of the 13 patients with RD, 10 were diagnosed as having benign childhood epilepsy with centro-temporal spikes (BCECTS), 2 as having Panayiotopoulos syndrome (PS), and 1 as having symptomatic or non-idiopathic localization-related epilepsy (LRE, unknown localization). Of the 17 patients with frontal PAs, 2 were diagnosed as having BCECTS, 1 as having PS, 9 as having frontal lobe epilepsy (FLE), and 5 as having LRE, unknown localization. One patient with occipital PA was diagnosed as having PS. Of three patients with a mid-temporal PA, 1 was diagnosed as having temporal lobe epilepsy, and 2 as having LRE, unknown localization. Of the 13 patients without an EEG PA, 2 were diagnosed as having benign partial epilepsy with a complex partial seizure, 1 as having FLE, 4 as having LRE, unknown localization, 5 as having GETCS, and 1 was undetermined. Among first EEG, 11 patients are useful for the diagnostic of idiopathic focal epilepsy such as BCECTS and PS.

4. Discussion

While all patients with epilepsy experience seizures, not all individuals with seizures have epilepsy. Thus, epileptic seizures must be clearly differentiated from other paroxysmal events, which have a different prognosis and imply a different therapeutic approach (Beghi, 2008). On the other hand, in many patients, seizures tend to remain isolated and, for this reason, drug treatment may not be justified, since drugs may cause significant adverse effects

Table 4
Associations between the localizations of EEG PAs and subsequent development of epilepsy.

Location of EEG PAs	Without epilepsy (n = 17)	With epilepsy (n = 35)	Odds ratio (95% CI) compared with generalized	p-Value
Generalized (n = 6)	5	1	n.a.	n.a.
Frontal (n = 18)	1	17	85.0 (4.5–1617.1)	0.0008
RD (n = 20)	7	13		0.065
Occipital (n = 3)	2	1		0.58
Mid-temporal (n = 5)	2	3		0.24

EEG, electroencephalogram; PA, paroxysmal abnormality; CI, confidence interval; n.a., not available.

(Mizorogi et al., 2015). For these reasons, risk factors of recurrence influencing the recommendations and decisions to treat someone who presents with a first unprovoked seizure need to be discussed.

Numerous studies have been performed to identify predictors of seizure recurrence. A meta-analysis by Berg and Shinnar identified an abnormal EEG and an underlying documented cause as the most consistent findings (Berg and Shinnar, 1991). More recently, a large, multinational, randomized trial (the MRC Multicentre trial for Early Epilepsy and Single Seizures; MESS) presented a risk allocation model for seizure recurrence using a regression model that included the presence of an abnormal EEG (Kim et al., 2006). Thus, EEG may have value in determining the prognosis of epileptic seizure recurrence in children.

The present study also showed that EEG findings had clinical value for predicting seizure recurrence after a first unprovoked seizure. Especially among them, the present results showed that the spike patterns in patients with seizure recurrence differed from those in patients without recurrence. Localization in the frontal region was revealed in 48.5% of patients with epilepsy, with only 2.8% in the generalized and occipital regions. In 94.4% of patients with frontal EEG PAs, epilepsy developed later. These findings suggest that the incidence of later development of epilepsy in patients with a first unprovoked seizure may depend on the EEG PA location. Patients with frontal EEG PAs may have a significantly higher risk of developing epilepsy than those with focal paroxysms in other regions of EEG foci.

Whether an EEG should be obtained routinely in patients with a first unprovoked seizure is still under debate. Beyond the effect on treatment, several factors that are intrinsic to the individual can influence the risk of recurrence. Several studies have reported that there are two factors that are consistently associated with an increased risk of recurrence in both children and adults (Berg and Shinnar, 1991; Ramos Lizana et al., 2000; Kho et al., 2006). One of these two factors is an abnormal EEG finding, particularly if the abnormality is an epileptiform PA. Moreover, from the data by Kim et al, patients with an abnormal EEG finding appear to have an even higher risk than those with only one of the factors (Kim et al., 2006). However, some authors would not treat after a first unprovoked seizure, even if the EEG is abnormal (Fountain and Freeman, 2006). The risk is that treatment will not be effective in preventing seizure relapse (Perucca, 2008). Despite these controversies over the value of EEG, EEG remains an essential tool for understanding the underlying process and providing the best counseling. The value of the EEG lies in three respects: suggesting the presence of focal lesions; allowing individual prediction of seizure recurrence; and indicating a specific epilepsy syndrome (e.g., 3-Hz spike-wave pattern) (Pohlmann-Eden and Newton, 2008). In the present study, a frontal EEG PA was identified as a significant risk factor for seizure recurrence as compared to paroxysms in other regions of EEG foci. Although the present sample was small, the data suggest that children with a first unprovoked seizure presenting with frontal paroxysmal EEG abnormalities may be at risk for the development of epilepsy later in life.

The reported yield of EEG PAs in routine examinations ranged from 12 to 27% (Hopkins et al., 1988; van Donselaar et al., 1992; Neufeld et al., 2000; Schreiner and Pohlmann-Eden, 2003), and it increased to 23–50% if sleep recordings could be obtained (van Donselaar et al., 1992; Schreiner and Pohlmann-Eden, 2003; King et al., 1988). The present results showed that the proportion with EEG PAs (59.8%) was greater compared to these previous reports. However, sleep recordings were obtained in the present study. In addition, the patients in this study were under 16 years of age. The proportion with epileptiform activity on the first EEG was found to be significantly greater in children compared to patients older than 16 years (Neufeld et al., 2000; King et al., 1988). The high rate of EEG PAs in the present study may be related to these

aspects. However, the present sample was too small to make definitive conclusions. Further investigations are needed to clarify these points.

A child who presents to the treating physician with a first unprovoked seizure faces a number of problems that require correct diagnostic and therapeutic management. Deciding when to begin treatment with AEDs in patients with a first seizure or infrequent seizures is difficult. Current evidence shows that the policy of immediate AED treatment, mainly carbamazepine or valproate sodium, in low doses reduces the occurrence of seizures in the next 1–2 years, but does not modify rates of longer-term remission in subjects with a single seizure (Chadwick, 2008). On the other hand, ongoing medical and psychosocial problems continue for patients with epilepsy, even in a sample where the majority report good control of their epilepsy (Jacoby, 1995; Fisher et al., 2000). In the present study, among first EEG, 11 patients (32.4%) are useful for the diagnostic of idiopathic focal epilepsy such as BCECTS and PS. The results of the present study may give additional information about the clinical factors for seizure recurrence. Moreover, this result could be also important to emphasize the role of sleep EEG after a first unprovoked seizure. Understanding these issues may improve communication of risks and benefits to individuals and may support making the decision to start or not to start AED treatment after a first seizure for clinicians and/or patients and their families.

Epileptic seizures may occur time dependently (Iasemidis et al., 1994). In the present study, all subjects were followed for more than 4 years. The duration of the seizure-free period that is necessary to consider the probability of a single seizure is difficult to determine. In most cases, the issue of whether a first seizure will remain isolated or whether it marks the onset of epilepsy is resolved within one or two years (Haut et al., 2007). In our previous study, the overall product-limit estimate of recurrence was 95.2% at 2 years after seizure onset (Mizorogi et al., 2015). Moreover, no recurrence was evident >4 years after the first seizure in that study. These findings suggest that the follow-up period in the present study may be appropriate to evaluate seizure recurrence.

The present study showed that the clinical profiles, such as age at onset, history of neonatal period, family history of epilepsy, and developmental delay prior to first seizure, were not associated with a greater risk of recurrent seizures. However, the present sample was too small to make any definite conclusions about these associations. Moreover, the methodology used in this study did not allow stratification for multiple factors and proper consideration of these associations. Accordingly, it is difficult to conclude that these clinical profiles are not associated with a greater risk of recurrent seizures from the results of this study. Further research is needed to evaluate these aspects.

In addition, although EEG findings may be important in children to predict subsequent epilepsy after a first unprovoked seizure, their significance can be improved when associated with imaging findings. However, patients with organic brain lesions on MRI were excluded in this study to avoid them affecting seizure recurrence. Accordingly, the association between EEG paroxysmal events and focal lesions observed on MRI could not be evaluated. Further investigations will clarify these points.

In conclusion, children with a first unprovoked seizure presenting with frontal paroxysmal EEG abnormalities may be at risk for the development of epilepsy later in life. We suggest that sleep EEGs should be a part of systematic comprehensive diagnostic approach in the patients with first unprovoked seizure. This finding could have a direct therapeutic impact of treatment (mostly not necessary) and parents' reassurance, especially in the idiopathic focal epilepsy such as BCECTS and PS. A careful workup including EEG is usually warranted. First unprovoked seizures are disruptive events for children and their families, and they

worry about seizure recurrence later in life. Adequate provision of information may reduce patient and parental fear.

Conflict of interest

We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines. None of the authors has any conflict of interest to disclose.

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